Optimizing Multiplier Performance Through Verilog Hardware Description Language Design

Wai Leong Pang
School of Engineering
Taylor's University
Subang Jaya, Malaysia
waileong.pang@taylors.edu.my

Tania Khatun
Faculty of Engineering
Multimedia University
Cyberjaya, Malaysia
tania72219@gmail.com

Swee King Phang
School of Engineering
Taylor's University
Subang Jaya, Malaysia
sweeking.phang@taylors.edu.my

Ajay Kumar Singh
Faculty of Electronics and
Communication Engineering
NIIT University
Rajasthan, India
Ajay.Singh@niituniversity.in

Angie See Tien Ng
NEX Network Edge Group
Intel Corporation,
Bayan Lepas, Malaysia
angie.see.tien.ng@intel.com

*Kah Yoong Chan
Faculty of Engineering
Multimedia University
Cyberjaya, Malaysia
kychan@mmu.edu.my
*Corresponding Author

***Abstract*— The Arithmetic Logic Unit (ALU) constitutes a critical component within a computer's Central Processing Unit, responsible for executing arithmetic and logical operations. Among its core elements, the multiplier holds particular significance, finding widespread use across various digital applications, notably in scientific computing and Digital Signal Processing. The pursuit of enhanced system performance necessitates the integration of a high-performance multiplier. The efficacy of the adder directly impacts the multiplier's efficiency. Since most of the time spent by a multiplier is on adding partial products, leading to considerable delays. This study offers a comprehensive evaluation of five commonly utilized adders—Ripple Carry Adder (RCA), Carry Save Adder (CSA), Carry Look-Ahead Adder (CLA), Carry Select Adder (CSLA), and Carry Skip Adder (CSKA)—across a 4-bit length to ascertain the most effective adder. The identified high-performance adder was subsequently employed in the design of three distinct 32-bit multipliers: Array, Booth, and Vedic. Utilizing the Verilog Hardware Description Language (HDL) for modeling, both adder and multiplier were simulated using Quartus Prime Lite. Performance comparison revolved around three key parameters: area, delay, and power consumption. Among the adders evaluated, the CLA adder exhibited approximately 15% less delay, alongside lower area and power consumption (1800 µm^2, 330 mW respectively) compared to its counterparts. Integration of the high-performance CLA adder into the Array multiplier showcased superior performance, evidenced by a significant reduction of approximately 50% in both area and delay, coupled with a notable 3.1% decrease in power dissipation compared to multipliers without CLA integration.**

Keywords*— *Carry Look-Ahead Adder, Array multiplier, Booth multiplier, Vedic multiplier, Verilog

I. Introduction

Multipliers stand as vital components in enhancing the velocity of digital systems, finding extensive application in numerous digital signal processing circuits and systems. Especially in signal processing applications and processors, high-performance multipliers play a pivotal role, facilitating crucial arithmetic operations like multiplication and addition [1]. The process of multiplication involves three distinct stages: partial product generation, reduction of partial products, and their subsequent addition. Improving multiplication efficiency typically involves reducing the number of generated partial products or enhancing their accumulation rate [2]. Given the critical role of the adder in multiplier performance, it becomes imperative to employ an efficient adder. A high-performance multiplier is characterized by low power consumption, rapid speed, and minimal area footprint [3].

This paper undertakes an evaluation of five commonly used adders—Ripple Carry Adder (RCA), Carry Save Adder (CSA), Carry Look-Ahead Adder (CLA), Carry Select Adder (CSLA), and Carry Skip Adder (CSKA)—to determine the most efficient adder for multiplier design. Further, we implement three widely employed multipliers—Array, Booth, and Vedic—for 32-bit operations, employing a high-performance adder. The evaluation criteria encompass resource utilization, area, delay, and power consumption. Simulation is conducted using Quartus Prime Lite 18.1, with Verilog HDL utilized for modeling both multipliers and adders.

Innovatively, we introduce a new 32x32 multiplier design by integrating Booth and Vedic multiplication structures. Utilizing a 16x16 Vedic architecture, the 32x32 multiplication process is optimized, significantly reducing the number of partial products generated. The Vedic architecture further enhances efficiency by partitioning signed inputs for rapid multiplication. The multiplier design incorporates carry-select and ripple-carry adders, aimed at efficiently adding the generated partial products. This implementation effectively minimizes propagation delay, contributing to overall performance enhancement. Functional design, simulation, and testing are carried out using Xilinx Integrated Synthesis Environment (ISE) and Vivado language. Comparative evaluation against 16x16 Vedic and Booth multipliers encompasses area, power consumption, and delay metrics [4].

The carry-save adder (CSA) plays a pivotal role in crafting 8-bit, 16-bit, and 32-bit iterations of the Vedic multiplier [5], strategically aimed at area optimization. A refined model was achieved by reconfiguring the architecture of the 4-bit Vedic multiplier, replacing conventional full adders with 2x2 binary multipliers and CSAs. Scaling up, the modified Vedic multiplier architecture was expanded to 32-bit through iterations of 8-bit, 16-bit, and 32-bit versions. Leveraging Verilog HDL, the multiplier's functionality was designed and synthesized, with simulation conducted using the ModelSim Altera simulator [5].

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A precursor study [6] mirrors the conceptual framework of this paper, introducing a modified binary Vedic multiplier employing Vedic sutras from ancient mathematics. Employing CSAs, multipliers spanning 4, 8, 16, and 32-bit were constructed, modeled using Verilog HDL. However, performance comparisons were limited to delay and logic utilization. The study concludes by suggesting the scalability of the proposed method to larger sizes and its potential application in diverse Carry Save Adder designs for modified Vedic multipliers [6].

VHDL serves as the modeling tool for evaluating the performance of the 32-bit array multiplier [7]. Employing two distinct adder types—carry look-ahead logic and carry-save adder—to process partial product terms, synthesis and simulation were employed to derive conclusive results. Notably, the carry-save adder demonstrates superior performance in power consumption, area efficiency, and delay, contrasting with the higher power consumption, delay, and area utilization associated with the carry look-ahead adder. FPGA Spartan III and Xilinx ISE facilitated the modeling and design of the multipliers.

Efficiency benchmarks for VLSI architectures emphasize minimal area, delay, and power consumption. A crucial component in digital multiplication processes, high-performance adders significantly influence a system's area, delay, and power characteristics. Consequently, evaluating and selecting optimal adders is imperative for constructing high-performance multipliers. This paper undertakes the comprehensive assessment of various adder types to pinpoint the most suitable candidate. Furthermore, it embarks on multiplier design integration with high-performance adders, aiming to minimize area, delay, and power consumption.

II. ADDERS

An adder circuit is indispensable to any digital system, serving as the fundamental operation in arithmetic functions. Its utility extends across various applications within modern VLSI systems, including the design of multipliers, ALUs, and different digital signal processing techniques like constructing Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters [8].

Within a multiplier, adders play a crucial role in summing the partial products generated during the multiplication process. To execute the multiplication operation, it necessitates breaking down the multiplication into a sequence of additions. Each partial product results from the multiplication of a single multiplier bit by the corresponding multiplicand bit, with the aggregated partial products yielding the final product. The choice of adder significantly influences the multiplier's performance [9]. The subsequent section delineates the performance of adders modeled in Verilog.

A. Ripple Carry Adder (RCA)

"The ripple carry adder is renowned for its elementary addition algorithms compared to other adders. Its size is determined by the number of full adders it comprises. Constructed through a cascade of full adders, the ripple carry adder integrates these connections. In this architecture, the first full adder block (FA) computes the addition of two bits along with a carry bit, yielding two outputs: a carryout and a sum. The carryout from one full adder feeds into the subsequent full adder, facilitating sequential computation. Consequently, this sequential transmission of carry results in increased delay as the size of the ripple carry adder expands [10]-[11]. Figure 2.1 illustrates the RTL view of a 4-bit ripple carry adder, modeled in Verilog.

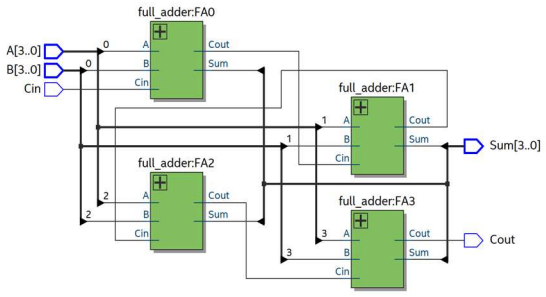


Figure 2.1: The RTL view of 4-bit ripple carry adder

B. Carry Save Adder (CSA)

The parallel counterpart of the ripple carry adder is recognized as the carry-save adder. This variant enables the addition of three or more binary numbers simultaneously. Operating on the same addition algorithm as the full adder (FA), the carry-save adder initially stores two inputs, sum, and carry, in the first stage before transferring them to the second stage. In the second stage, acting akin to a ripple carry adder (RCA), the saved sum and carry bits are added. However, instead of propagating to the carry-in of the subsequent full adder, each full adder's carry and sum bits are preserved and added to the subsequent sum term via a ripple carry adder [11]-[12]. Figure 2.2 depicts the RTL view of the carry-save adder for various bit configurations.

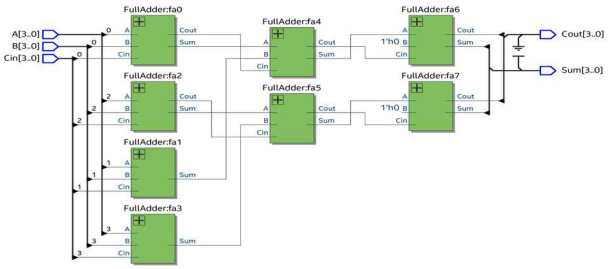


Figure 2.2: The RTL view of 4-bit carry save adder

C. Carry Look Ahead Adder (CLA)

The carry-lookahead adder (CLA) comprises two key components: the carry-generate and carry-propagate units. Renowned for its speed, the CLA outpaces other adder types by significantly reducing computation time for carry bits. It achieves this by continuously computing sum and carry bits through separate carry generation units. While carry propagation generates output via OR operation of two inputs, carry generation employs AND operation of two inputs, simultaneously executed. The carry and propagation bits from the preceding stage utilize the XOR operation to compute the sum bit [12]. Below is the RTL view of the CLA adder for different bit configurations.

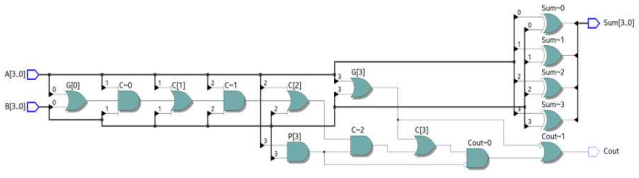


Figure 2.3: The RTL view of 4-bit carry look-ahead adder.

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D. *Carry Select adder (CSLA)*

The carry-select adder (CSLA) is built by combining a ripple carry adder and a multiplexer. Employing multiple ripple carry adders alongside a multiplexer, the CSLA adder selects the appropriate sum and carry bits from the correct adder based on the carry-in, which is determined at the outset of computation. This innovative approach effectively minimizes propagation delays and accelerates addition processes within digital circuits [12]-[13]. Below, you'll find the RTL view of the CSLA adder for various bit configurations.

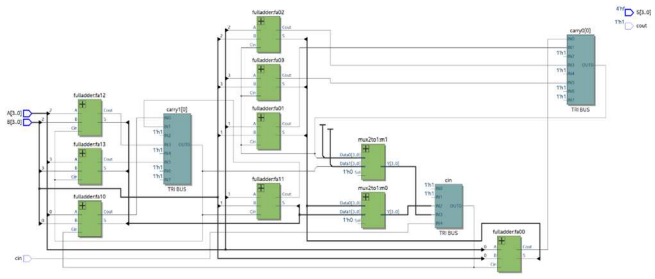


Figure 2.4: The RTL view of 4-bit carry select adder

E. *Carry Skip Adder (CSKA)*

The carry-skip adder, also referred to as a carry-bypass adder, offers a streamlined approach to enhancing the latency of a ripple carry adder (RCA) when compared to other adders. This efficiency is achieved by preemptively selecting the preceding carry without waiting for the carry generated by the RCA block, facilitated by the utilization of a 2-to-1 multiplexer as the carry skip logic [13]. The RTL simulation of the carry-skip adder is presented below the figure.

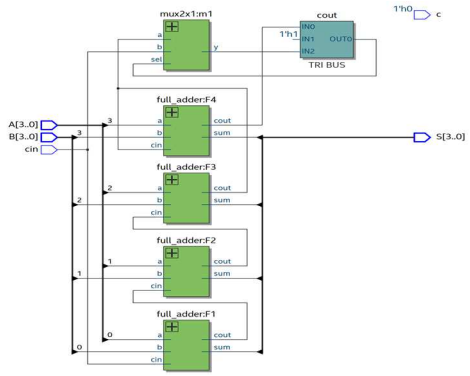


Figure 2.5: The RTL view of 4-bit carry skip adder

F. *Performance Analyses of the Adders Modeled*

This section showcases the simulation outcomes of adders across various bit lengths, modeled using Verilog Hardware Description Language. Five distinct types of 4-bit adders underwent simulation. The ensuing table illustrates the resource utilization of these different 4-bit adders. Furthermore, Figures 2.6, 2.7, and 2.8 depict the delay, area consumption, and power dissipation, respectively, of adders across 4, 8, and 16-bit configurations.

As illustrated in Figures 2.6, 2.7, and 2.8, the 16-bit RCA exhibits the poorest performance, characterized by the highest delay (17.942 ns), area consumption (2199 μm^2), and power dissipation (365.44 mW) among all the adders modeled in Verilog. Conversely, Figure 2.6 demonstrates that the CLA adder boasts the lowest delay, registering a 24.1% reduction compared to the RCA. Moreover, the carry skip adder follows closely with the second-lowest delay, presenting a 17.2% improvement over the RCA. Notably, the CLA adder also showcases the lowest area consumption and power dissipation compared to other adders. Specifically, the 16-bit CLA incurs a 17.6% decrease in area consumption (1813 μm^2) and a 5.4% reduction in power dissipation (345.78 mW) relative to the RCA. Similarly, the CSKA achieves the second-lowest area consumption and power dissipation, with reductions of 9.8% and 1.7% respectively compared to the RCA.

In summary, five adders were successfully modeled in Verilog, with the CLA adder emerging as the top performer due to its lowest delay, minimal area consumption, and reduced power dissipation. The next section will model the multipliers (Array, Booth, and Vedic) using the CLA adder to determine the best-performing multiplier in Verilog.

TABLE I. RESOURCES CONSUMPTION OF VARIOUS 4-BIT ADDERS

Adder	Total Adaptive Look-up Table (ALUT)	Logic Utilization
Ripple Carry Adder (RCA)	4	4
Carry Save Adder (CSA)	5	5
Carry Look Ahead Adder (CLA)	3	3
Carry Select Adder (CSLA)	22	19
Carry Skip Adder (CSKA)	28	27

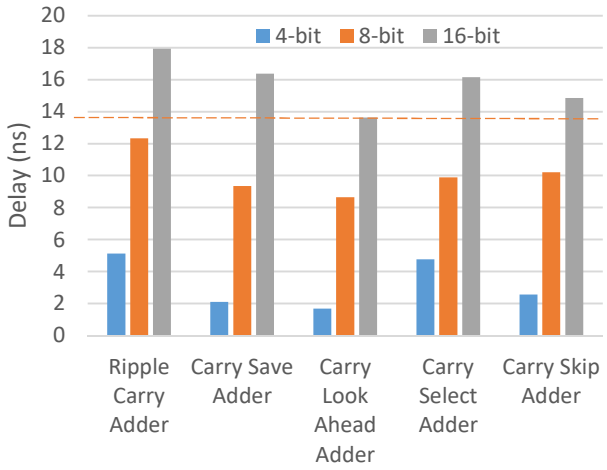


Figure 2.6: The delay of various adders

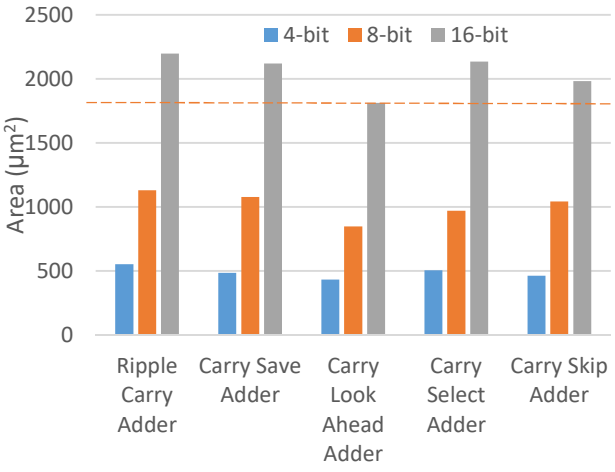


Figure 2.6: The area consumption of various adders

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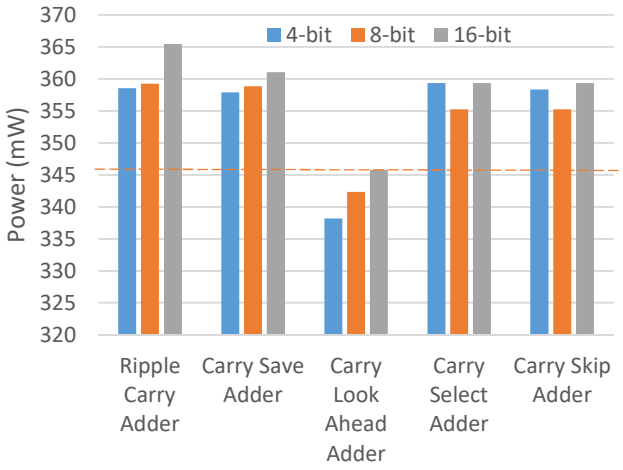


Figure 2.6: The power dissipation of various adders

III. MULTIPLIER

Multipliers play a pivotal role in a multitude of digital applications, particularly in high-performance devices. Drawing from extensive research, this paper focuses on three commonly utilized multipliers: Array, Booth, and Vedic multipliers. These variants have demonstrated superior performance across various digital applications, making them prime choices for investigation. Through comprehensive simulations, we meticulously evaluate the performance of these 32-bit multipliers.

A. Array Multiplier

The digital combinational circuit array multiplier employs an array of full and half adders to multiply two binary values. This multiplier functions via the add-and-shift operation, where the AND logic generates partial product bits by combining the multiplier with the multiplicand. Subsequently, the result undergoes leftward shifting by an appropriate number of bits. To obtain the total partial products, each partial product is added to the previously accumulated partial product. However, this method of combining partial products typically results in high power consumption and delay.

To mitigate these limitations, the use of a carry-lookahead adder (CLA) presents a viable solution. While the fundamental algorithm of the multiplication process remains unchanged, the addition of partial and final products differs slightly when employing the carry-lookahead adder. This addition method allows for lookahead computation of carry bits, thereby reducing overall power consumption and delay [14]-[15].

Incorporating a carry-lookahead adder (CLA) within an array multiplier yields significant improvements in area, delay, and power utilization. By minimizing the number of gates required to compute carry bits, the CLA enhances area utilization and effectively reduces the total area of the multiplier [16]. Figure 3.1 illustrates the RTL view of the modeled array multiplier utilizing a CLA.

B. Booth Multiplier

The Booth multiplier represents an innovative approach to multiplication, allowing for the simultaneous addition of three bits and a reduction in the number of partial products. Leveraging Booth encoding facilitates the concurrent execution of multiple multiplication steps, thereby expediting the multiplication process. Employing an adder-subtractor, Booth's method algorithm achieves comparable speed and compactness to a simple adder, a notable advantage.

By identifying three consecutive bits with identical values, Booth encoding effectively eliminates the need for addition and subtraction operations in certain cases. Consequently, Booth Multiplication often exhibits lower delay compared to other multiplier techniques. By examining three bits at a time, the Booth multiplier reduces the number of required adders, thereby reducing the time needed to compute partial sums. However, despite its exceptional efficiency, the Booth multiplier does entail a drawback in power consumption. This is primarily due to the requirement of multiple adder cells, which consume a significant amount of power [17]-[18]. The RTL view of the Booth Multiplier with a CLA adder is presented in Figure 3.2.

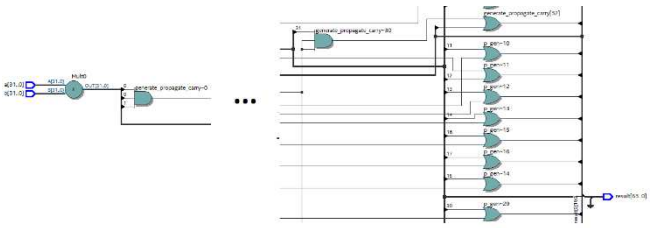


Figure 3.1: The RTL view of array multiplier using CLA

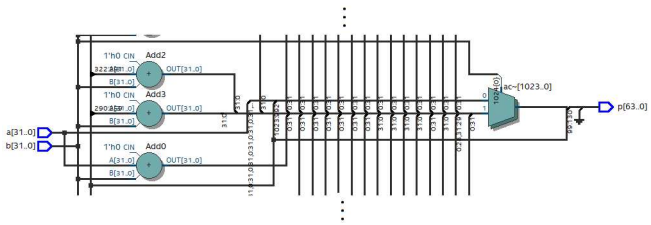


Figure 3.2: The RTL view of booth multiplier using CLA

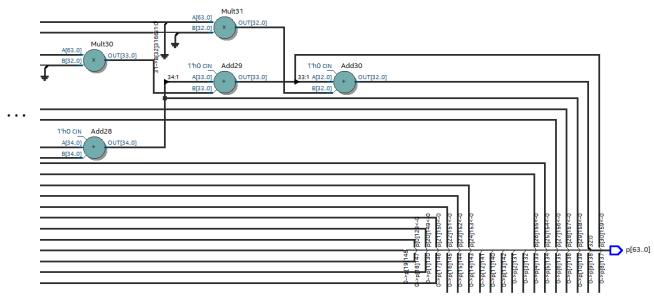


Figure 3.3: The RTL view of Vedic multiplier with CLA

C. Vedic Multiplier

Vedic mathematics, an ancient mathematical system originating from India, offers direct applications in algebra, arithmetic, and various mathematical domains. By streamlining calculation processes and eliminating unnecessary stages, it significantly reduces complexity. The efficient and high-speed Vedic multiplier, built upon the principles of Vedic mathematics, utilizes specific algorithms known as "sutras" to execute multiplication operations. Notably, this multiplier excels in reducing propagation delay, optimizing power consumption, and enhancing multiplication speed within digital systems.

Integrating the Vedic multiplier can be achieved through the utilization of a Carry Look Ahead Adder (CLA) as a pivotal component for swift addition. This approach involves pre-calculating carry signals from input signals using the

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CLA, thereby mitigating delays encountered during the addition process. Moreover, employing the CLA adder contributes to reduced power consumption and area utilization. Figure 3.3 illustrates the RTL view of the Vedic multiplier, utilizing the Carry Look Ahead Adder (CLA) to predict steps based on input signals. Rooted in ancient Vedic mathematics, the design emphasizes vertical and crosswise techniques. By generating partial products and sums in a single step, the multiplier streamlines complexity and enhances operational speed.

D. Simulation Results

Three performance metrics—area consumption, delay, and power dissipation—are employed to compare the performance of existing multipliers without Carry Look Ahead (CLA) and proposed multipliers with CLA. The multipliers are modeled using Verilog. Figure 3.4 illustrates the area consumption of the multipliers with and without CLA. Notably, the array multiplier with CLA exhibits the lowest area consumption among the compared variants. Specifically, the area consumption of the array multiplier with CLA (1168 μm^2) is 51.2% lower than that of the array multiplier without CLA (1766 μm^2). Additionally, the area consumption of the Booth multiplier and Vedic multiplier with CLA is 12.7% and 7.9% lower, respectively, than their counterparts without CLA. This highlights the significant reduction in area consumption achieved by integrating CLA into the modeled multipliers.

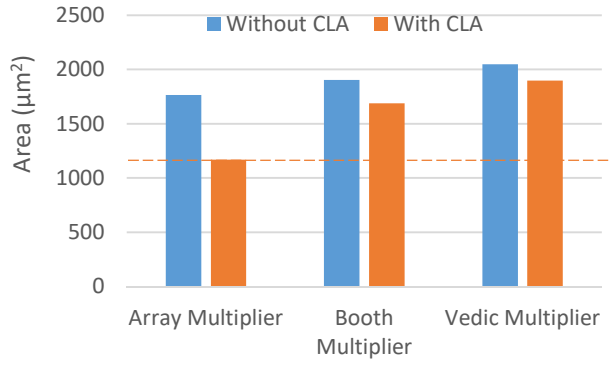


Figure 3.4: The area consumption of the multipliers modeled

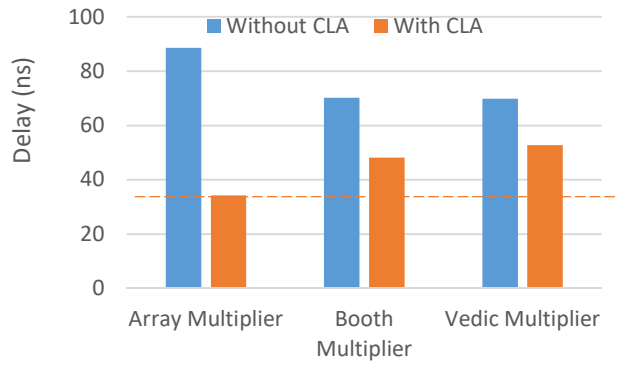


Figure 3.5: The delay of the multipliers modeled

Figure 3.5 depicts the delay of the multipliers with and without CLA. Once again, the array multiplier with CLA showcases the lowest delay among the compared configurations. Specifically, the delay of the array multiplier with CLA (34.26 ns) is 158.5% lower than that of the array multiplier without CLA (88.56 ns). Similarly, the delay of the Booth multiplier and Vedic multiplier with CLA is 45.6% and 32.4% lower, respectively, than their counterparts without CLA. This underscores the significant reduction in delay achieved by incorporating CLA into the modeled multipliers.

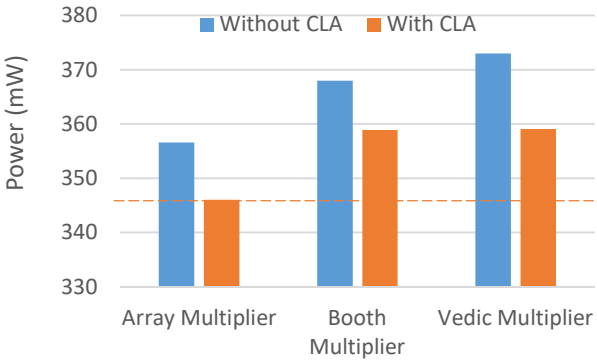


Figure 3.6: The power dissipation of the multipliers modeled

Figure 3.6 illustrates the power dissipation of the multipliers with and without Carry Look Ahead (CLA). Notably, the array multiplier with CLA exhibits the lowest power dissipation among the compared variants. Specifically, the power dissipation of the array multiplier with CLA (346.03 mW) is 3.1% lower than that of the array multiplier without CLA (356.61 mW). Furthermore, the power dissipation of the Booth multiplier and Vedic multiplier with CLA is 2.5% and 3.9% lower, respectively, than their counterparts without CLA. This underscores the significant reduction in power dissipation achieved by integrating CLA into the modeled multipliers.

The performance analysis reveals that multipliers with CLA require less time to generate final results compared to existing multipliers without CLA. All three multipliers demonstrate lower delay, reduced area, and decreased power dissipation compared to the existing models. Notably, the proposed model with CLA indicates that the array multiplier outperforms the Booth and Vedic multipliers. The array multiplier with CLA exhibits a remarkable reduction of more than 50% in delay and area utilization compared to the existing multiplier. The integration of CLA enhances performance by reducing the number of gates required to calculate carry bits, thereby decreasing the time needed to compute the result of the more significant value bits of the adder. Consequently, there is a notable improvement in power consumption, delay, and area utilization.

IV. Conclusion

In this paper, we modeled five adders—RCA, CSA, CLA, CSLA, and CSKA—using Verilog for 4-bit, 8-bit, and 16-bit configurations to identify the best performer for multiplication operations. The CLA adder stood out with the lowest delay, area consumption, and power dissipation, while the RCA performed the worst. Specifically, the 16-bit CLA showed improvements of 24.1% in delay, 17.6% in area consumption, and 5.4% in power dissipation compared to the 16-bit RCA. The 32-bit Array, Booth, and Vedic multipliers with CLA are modeled. Simulations revealed that the Array multiplier with CLA integration had the best performance, with a 51.2% reduction in area, a 158.5% decrease in delay, and a 3.1% reduction in power dissipation compared to the non-CLA version. The CLA also improved the performance of Booth and Vedic multipliers. Future work could expand to include more types of adders and multipliers, higher bit lengths, and

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comparisons with the existing adders designed at the transistor level.

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